

Element A

Questions for Mrs. Brennan

The Life Skills classroom at Whitemarsh Elementary has hired the PW engineering class to design and fabricate a new toy which combines learning with stimulating components such as lights, sounds, and colors. The class harbors learning around skills used in everyday life, skills that would seem like basic tasks to others, but are difficult for students with disabilities. These may vary between physical and mental disabilities. Neurodivergent is a framework for understanding human brain function and mental illness. It argues that diversity in human cognition is normal and that some conditions classified as mental disorders are differences and disabilities that are not necessarily pathological. Sensory cause and effect toys provide a source of vital enrichment for neurodivergent kids, encouraging them to engage with their environment in new ways. The idea of cause and effect, or action and reaction is a type of relationship between actions the teachers would like for the students to learn. The Life Skills class has few ways to teach the students the idea of cause and effect, and would like more. The class has 8 students with varying skill and knowledge levels, so the toy must be able to work for all of them and be cleaned/disinfected after use.

The students in the life skills classroom currently have only a few sensory toys, and a few cause and effect toys. These toys are stimulating for the kids and often work on motor skills, but are not always educational. Our goal is to create a cause and effect toy that effectively combines both sensory stimulation and learning, making a unique addition to the tools that the teachers use with the students. The kids stay much more engaged with these sensory toys, and having one that will simultaneously teach them will be an effective tool for the teachers to use. It is an item that could be used throughout the day in this classroom, so it would be an invaluable item. The group emailed the head teacher, Mrs. Brennan, a list of questions to help us identify the problem and have a better understanding of what a solution would look like. We also asked for a list of the sensory toys the classroom already has, so we can design something to make up for what they don't have.

This project is very flexible in that it is applicable to multiple contexts. Sensory toys and cause and effect toys are very good for kids with autism because the stimulation coupled with the ability to influence their environment is appealing for them. However young children still in early developmental stages find sensory toys and cause and effect toys appealing as well. Most toys for young toddlers and children incorporate aspects of both sensory and cause and effect. Because of this our design would be fitting for any pre school, elementary classrooms, or homes with young children as well. Everything in the world is the product of a cause and effect relationship. This toy can help all kids understand that their actions will produce results. It allows them to expand on their current knowledge of the world and helps them with *life skills*.

The primary stakeholders of this project are the life skills students. Our project is based on them and their needs, which are communicated through the head of the classroom Mrs. Brennan, as they are unable to convey them to us. Many of our design constraints and requirements came from our conversations with her about our project. Our secondary stakeholders in this project are the aids, nurses, and teachers that are working with the students in the classroom. As they are directly interacting with the students, their daily routines and schedules would be impacted by the students' use of the toy. It would

also provide a new resource for them to use with the students in the classroom. The tertiary stakeholders consist of the parents of the Life Skills class students as this toy will be helping them with their motor skills and developmental learning. It may impact how they are acting at home and with their parents as these skills grow. Finally the quaternary stakeholders are the principles allowing us to work with the class. As the sponsors of our projects, our project is being allowed by them so the success of it would impact them. Potentially if the project is successful, our product could be used in other classes in the district or other life skills classes in other schools. That would all be coordinated through the principle.

- ❖ <https://www.sensoryfriendly.net/what-is-a-sensory-friendly-environment/>
 - This citation covers the effects that noise, lighting, seating, scents, and clutter have on a sensory friendly environment. The benefits of this environment help individuals who are easily overstimulated that normally would experience a high-stress response and discomfort if they are in loud environments.
- ❖ Verbal Expert: John Staudenmayer
 - Discussing the effects between reinforcement and punishment. Reinforcement promotes behavior regardless of the fact whether it is positive or negative. (taking away an unwanted stimulus/providing a favorable stimulus)
- ❖ [https://www.churchillstl.org/learning-disability-resources/nonverbal-learning-disorder/#:~:text=N%20onverbal%20learning%20disorder%20\(NVLD\)%20is.and%20comprehension%20of%20abstract%20concepts](https://www.churchillstl.org/learning-disability-resources/nonverbal-learning-disorder/#:~:text=N%20onverbal%20learning%20disorder%20(NVLD)%20is.and%20comprehension%20of%20abstract%20concepts)
 - This citation covers the nonverbal aspects that individuals with disabilities may face. Children with nonverbal tendencies are known to be physically clumsy, often bumping into objects or people. They may have trouble communicating with others, especially with conversations of their desires. Reading, mathematics, metaphors or abstract concepts are hard to understand and grasp.
- ❖ https://www.handyhandouts.com/viewHandout.aspx?hh_number=486&nfp_title=Impact+of+Colors#:~:text=Research%20confirms%20that%20color%20affects.hyperactive%20or%20bored%20and%20uninterested
 - This citation covers how colors can impact the way children feel and act. Color affects learning, communication, productivity, and emotions. This causes both positive and negative impacts on children. Color selection can cause children to become excitable and hyperactive or bored and uninterested
- ❖ <https://www.soundproofcow.com/how-sound-affects-your-childs-development/#:~:text=Though%20sounds%20such%20as%20music.attention%20and%20problem%2Dsolving%20skills>
 - This citation covers how sounds can affect children's learning skills. Studies have found that children exposed to consistent noise have more trouble completing school work. This noise influences their reading attention and problem-solving skills.
- ❖ <https://www.hightspeedtraining.co.uk/hub/how-to-support-a-child-with-autism-in-the-classroom/>
 - This citation covers how to properly support children on the spectrum in the classroom. A poor classroom environment for special individuals can hugely disadvantage them. It can cause them difficulty with engaging in learning activities and coping with daily life. By integrating suitable learning styles and alleviating any discomfort in the classroom, you

will enable children to take part in learning more comfortably and become better prepared for their future.

Whitemarsh Elementary's life skill's class consists of 8 young students who require extra assistance from specialized personnel. There are different types of stimulation that can impact a child's learning habits while also allowing them to learn a foundation for further knowledge. Different rewards offered from these instruments include: sounds, lights, and vibration. It was found that when introducing rewards into a learning environment, performance increases by 27%. Upon entering the classroom there are a limited number of devices that fit the category of what we are creating. Around 90% of teachers in America are required to purchase their own materials with an average of \$673 per year for their classrooms. Our design will be innovative and created without any cost to the teacher.